

Ross Baker

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Education

University of Oxford – MMath Mathematics (Integrated Master's)

Oct 2022 – Jun 2026

Master's Distinction; Undergraduate 2:1

Theories of Deep Learning
Information Theory
Optimal Control
Computational Game Theory

Algorithmic Foundations of Learning
Advanced Topics in Statistical Machine Learning
Optimisation for Data Science
Probability, Measure and Martingales

Dissertation – Beyond Integrated Information Theory: A Formal Framework for Emergent Causation and the Foundations of Consciousness [🔗](#)

Research Outputs

Temporal Task Diversity: Inductive Biases Under Non-Stationarity in Synthetic Sequence Modelling [🔗](#)

Nov 2025 – May 2026

Accepted to the ICML 2026 Workshop on Foundations of Deep Generative Models; joint first author

- Extended the in-context linear regression framework of Raventós et al. (2023) to non-stationary task distributions, motivated by settings such as curriculum learning, multi-stage training, and on-policy reinforcement learning
- Showed that increasing non-stationarity lowers the task-diversity threshold, biasing transformers toward generalisation over memorisation

Optimal Affine Activation Steering Methods for Unlearning [🔗](#)

Dec 2025 – Jan 2026

Accepted to TAIS 2026, the Technical AI Safety Conference (non-archival); sole author

- Unified existing activation steering methods under a common optimisation framework, characterising each as the solution to a specific constrained objective
- Proposed three unlearning methods, each derived by minimising an objective designed to penalise disruption to untargeted knowledge
- Evaluated each method on transformers trained on Markov-chain mixtures, observing targeted removal of memorised patterns with no measurable effect on in-context inference, providing preliminary evidence of knowledge removal rather than mere output suppression

Short Courses & Reading Groups

Oxford AI Safety Initiative – Alignment Research Bootcamp

2 weeks full-time, Jun – Jul 2026

Completed a compressed ARENA syllabus; concluded with an ongoing group research project

- Implemented methods across transformer interpretability, reinforcement learning, LLM evaluations, and alignment science
- Reproduced and built on *Inoculation Prompting* (Wichers et al., 2025) to test the Persona Selection Model, extracting persona vectors and running alignment evals across diverse domains to assess whether misalignment generalises as a persona-level trait

Oxford AI Safety Initiative – AI Strategy Series

Nov 2025

Explored timelines, threat models, and the geopolitics of transformative AI; concluded with a wargame

Oxford AI Safety Initiative – Singular Learning Theory

Nov 2025 – Dec 2025

Studied selected chapters of Algebraic Geometry and Statistical Learning Theory (Watanabe)

Effective Altruism Oxford – Moral Uncertainty

Oct 2025 – Nov 2025

Studied frameworks for decision making under normative uncertainty

HarvardX – CS50's Introduction to Programming with Python

9 weeks part-time, Jul – Aug 2025

Research Interests

Representation engineering – affine activation steering; distinguishing knowledge removal from output suppression in unlearning

In-context learning – transitions from memorisation to generalisation; inductive biases under non-stationarity

Emergent misalignment – inoculation prompting; the Persona Selection Model

Mathematical consciousness science – emergent causation in stochastic systems; critiques of Integrated Information Theory

Employment

Freelance Jazz Drummer [🔗](#)

part-time, 2022 – present

- Perform professionally in modern jazz small groups, currently in the Alexander McNamee Quartet
- Released albums with the Oxford University Jazz Orchestra and the jazz-fusion band OCULAR
- Extensive experience collaborating and performing under pressure

Hertford College International Programmes – Residential Assistant

13 weeks full-time, summers 2024 – 2026

- Supported student cohorts from China and Japan during residential programmes
- Led team-building activities and educational discussions